



Levers

A lever is a bar that can turn around a fixed point (fulcrum). If you apply a force (effort) to one part of a lever, another part exerts a force (load).



One type of lever works like a seesaw with the fulcrum between the load and the effort.



Another type places the load between the fulcrum and the effort (as on a wheelbarrow).



A **third type** of lever, shown by tongs, places the effort between the fulcrum and the load.

Machines

Machines make tasks easier. They reduce the effort you need to move something or the time that it takes. They work either by spreading the load or by concentrating your efforts. All the machines you see here are called simple machines.



Wheel and axle

An axle goes through the center of a wheel. Together they work as a simple rotating machine that makes it easier to move something from one place to another.



Gears

Gears are wheels with teeth that interlock so that one wheel turns another. They increase speed or force. Gears on a bicycle affect how much you must turn the pedal to spin the wheel.



The pedal turns a wheel, which turns a smaller wheel at a greater speed.



hands on

Try walking straight up a hill and then zigzag your way up. The winding path works like a simple machine. It increases the distance you walk, but decreases the effort you use.